Open the evidence document, **evidence.doc**

Make sure that your name, centre number and candidate number will appear on every page of this document. This document must contain your answers to each question.

Save this evidence document in your work area as:

evidence_ followed by your centre number_candidate number, for example: **evidence_zz999_999**

A class declaration can be used to declare a record.

If the programming language does not support arrays, a list can be used instead.

A source file is used to answer **Question 1**. The file is called **`IntegerData.txt`**

1 The text file `IntegerData.txt` stores 100 integer numbers between 1 and 100 inclusive. A program is required to read in this data and perform searching and sorting on the data.

(a) Write program code to declare a global 1D array, `DataArray`, with space for 100 integer values.

Save your program as **Question1_N22**. Copy and paste the program code into **part 1(a)** in the evidence document.

[2]

(b) The procedure `ReadFile()` must read in the numbers from the text file and store each one in the array. Use appropriate exception handling.

Write program code for the procedure `ReadFile()`.

Save your program. Copy and paste the program code into **part 1(b)** in the evidence document.

[6]

(c) The function `FindValues()` asks the user to enter a number to search for in the array. The number input must be a whole number between 1 and 100 inclusive. The function then returns the number of times the number input appears in the array.

Write program code for the function `FindValues()`.

Save your program. Copy and paste the program code into **part 1(c)** in the evidence document.

[7]

(d) **(i)** Write program code to call `ReadFile()` and `FindValues()` from the main program. The return value from `FindValues()` must be output with an appropriate message.

Save your program. Copy and paste the program code into **part 1(d)(i)** in the evidence document.

[3]

(ii) Test your program using the number 61 as input.

Take a screenshot to show the output.

Copy and paste the screenshot into **part 1(d)(ii)** in the evidence document.

[1]

(e) The procedure `BubbleSort()` needs to perform a bubble sort on the array and print the contents of the sorted array.

Write program code for the procedure `BubbleSort()` **and** call it from the main program.

Save your program. Copy and paste the program code into **part 1(e)** in the evidence document.

[4]